Fiscal Year:	FY 2007	**Task Last Updated:**	FY 02/04/2008
PI Name:	Kimbrell, Deborah A Ph.D.		
Project Title:	Drosophila as a model of immune function in conditions of altered gravitational force		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology (3) Animal Biology: Invertebrate		
Space Biology Cross-Element Discipline:	(1) Immunology		
Space Biology Special Category:	None		
PI Email:	dakimbrell@ucdavis.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	530-752-1116
Organization Name:	University of California - Davis		
PI Address 1:	Molecular and Cellular Biology		
PI Address 2:			
PI Web Page:			
City:	Davis	**State:**	CA
Zip Code:	95616	**Congressional District:**	25
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2003 Space Biology 03-OBPR-03
Start Date:	03/01/2004	**End Date:**	09/30/2007
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	Shuttle		
Flight Assignment:	STS-121 NCE through 9/30/2007, per A. Chu (8/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNA04CC76A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Immune function is essential for survival and involves basic cellular processes such as non-self recognition, cell growth and apoptosis. Immune functions are known to be affected under conditions of altered gravity. However, how this translates to disease versus health, and normal versus cancerous cell growth is not understood. The most fundamental part of immune function is innate immunity, and this is reflected by the fact that innate immunity is a phylogenetically ancient system. The main studies that have revealed the similarities in the immune systems of vertebrates and invertebrates are those that compare the fruit fly Drosophila and mammals. In particular, many components of the signal transduction cascades involved in the response to infection are conserved in flies and humans. In Drosophila, this results in the induction of antimicrobial peptides, the hallmark of the humoral response. Similarly, conserved pathways in the cellular system are involved in immune cell growth and regulation. This project is designed to use the power of Drosophila as a model of innate immunity to investigate the effects of gravity on immune function. A key advantage of this approach is the ability to compare normal flies to mutant flies that are either defective in an aspect of immune function or defective in gravitational response.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This work will contribute toward an understanding of the fundamental innate immune response and thus help identify the parameters that are important for health and targets that could be useful for disease treatment.
Task Progress:	We have found that immune function in Drosophila is altered in conditions of hypergravity compared to normal gravity. Specifically, flies show a significant increase ($P < 0.05$) in survival rates at 4g versus 1g after infection with the fungus Beauveria bassiana. An increase in survival rate is consistent with either the infecting organism being weaker and/or the flies being stronger in response. By comparing these results to gravity-sensing mutants, the relative roles of gravity, host and pathogen can begin to be discerned. We thus tested yuri mutants, named after the Russian astronaut Yuri Gagarin and isolated as having a defective gravitaxis response, for their survival after infection and exposure to 4g hypergravity. The UAS-yuri rescue stock, which restores normal gravitaxis to yuri mutants, showed the same increase in survival as the other stocks tested. The yuri mutants, however, showed no change in survival at 4g compared to 1g. Thus, the data clearly indicate that gravity has an effect on the host response. This work was reported in the abstract, “Taylor, K. A., Kloehn, A. T, Fuller, P. M., Beckingham, K. M., Fuller, C. A. and Kimbrell, D. A., Alteration of gravitational force affects the immune response, 46th Annual Drosophila Research Conference, 2006,” and is being followed up by a corresponding manuscript. These results with hypergravity and the yuri mutants made it clear that it is important to understand how gravity is sensed. Preliminary experiments related to determining how gravity is sensed have been published, “Syed, Z., Ishida, Y., Taylor, K., Kimbrell, D. A. and Leal, W. S. (2006) Pheromone reception in fruit flies expressing a moth's odorant receptor. Proceedings of the National Academy of Sciences USA 103:16538-16543.” In addition, we have found that the survival of infected flies is very different for male versus female flies. Generally, only male flies have been tested in immune response studies, and those that have tested females have addressed only specific features. Our studies have shown sex-specific differences in post-infection survival ($P<0.01$). This work is now published, “Taylor, K. and Kimbrell, D. A. (2007) Host immune response and differential survival of the sexes in Drosophila. Fly 1: 197-204.” A manuscript based on experiments to analyze blood cell melanotic tumor formation in Drosophila is in preparation. All of these studies have also supported our FIT (Fungus, Immunity and Tumorigenesis) Flight Experiment, funded as grant NNA05CV0A, which flew successfully on the space shuttle Discovery in July 2006. Altogether, our experiments have shown significant changes in the immune response in altered gravitational forces. However, because of NASA’s budget cuts and virtual elimination of basic Life Sciences funding, these results cannot be completed and the new directions that these results have revealed cannot be followed up.
Bibliography Type:	Description: (Last Updated: 07/07/2015)
Articles in Peer-reviewed Journals	Taylor K, Kimbrell DA. "Host immune response and differential survival of the sexes in Drosophila." Fly 2007 Jul-Aug;1(4):197-204. https://www.landesbioscience.com/journals/fly/article/5082/ , Aug-2007
Articles in Peer-reviewed Journals	Syed Z, Ishida Y, Taylor K, Kimbrell DA, Leal WS. "Pheromone reception in fruit flies expressing a moth's odorant receptor." Proceedings of the National Academy of Sciences USA. 2006 Oct 31;103(44):16538-43. PMID: 17060610 , Oct-2006
Articles in Peer-reviewed Journals	Kleinhesselink K, Conway C, Sholer D, Huang I, Kimbrell DA. "Regulation of hemocytes in Drosophila requires dappled cytochrome b5." Biochem Genet. 2011 Jun;49(5-6):329-51. Epub 2011 Jan 30. http://dx.doi.org/10.1007/s10528-010-9411-7 ; PubMed PMID: 21279680 , Jun-2011
Articles in Peer-reviewed Journals	Taylor K, Kleinhesselink K, George MD, Morgan R, Smallwood T, Hammonds AS, Fuller PM, Saelao P, Alley J, Gibbs AG, Hoshizaki DK, von Kalm L, Fuller CA, Beckingham KM, Kimbrell DA. "Toll mediated infection response is altered by gravity and spaceflight in Drosophila." PLoS One. 2014 Jan 24;9(1):e86485. http://dx.doi.org/10.1371/journal.pone.0086485 eCollection 2014. PubMed PMID: 24475130 , Jan-2014